

Transient Noise in ngspice

Built-in sources and OSDI (Verilog-A) devices: mechanism, amplitude law, and verification

Ngspice + OpenVAF Enhancements project

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Transient Noise in ngspice — built-in sources and OSDI (Verilog-A) devices

Small-signal `.noise` linearises about an operating point and reports spectral densities. That is the right tool for an amplifier's input-referred noise, and the wrong tool for anything where the noise itself changes the trajectory: jitter, noise-induced switching, comparator metastability, an oscillator's phase noise. Those need the noise present in the time domain, inside the Newton loop, so that the circuit responds to it.

This note describes how ngspice does that — the built-in `trnoise` sources it has had for years, and the OSDI path added by [Enhancement-364](#) that makes Verilog-A devices noisy in `.tran`. It then derives the amplitude law from the generator's structure and checks it against closed-form results.

Everything quoted below is measured, not asserted. The figures are produced by [make_trnoise_figs.py](#), which writes the Verilog-A, compiles it with `openvaf-r`, runs ngspice and plots the result against the analytic curve.

1. What is and is not a noise source in `.tran`

This is the first thing that surprises people, and it is worth stating plainly before any formula:

element	noisy in .noise	noisy in .tran
built-in resistor R	yes ($4kT/R$)	no
built-in diode / BJT / MOS	yes	no
independent source with <code>trnoise(...)</code>	no (it is a source)	yes
OSDI device with <code>white_noise/flicker_noise</code>	yes	yes, since E-364

A plain R contributes thermal noise to a .noise sweep and contributes nothing to a transient. That is not an oversight to route around; it is what makes transient noise affordable, since otherwise every resistor in a netlist would need its own random stream. It does mean that a transient measurement and a .noise measurement of the *same circuit* are only comparable when the resistors are either noiseless in both or negligible in both — a trap that section 6 shows biting a real measurement.

Verified directly. The circuit below has a $1\text{ k}\Omega$ resistor and a 1 nF capacitor; if the resistor were noisy the variance across C would be $kT/C = 4.14\text{ pV}^2$:

```
builtin resistor is noiseless in tran
V1 in 0 dc 0
R1 in mid 1k
C1 mid 0 1n
Vn nz 0 dc 0 trnoise(0 1e-8 0 0)      $ activates transient noise
Rz nz 0 1k
.control
tran 1e-8 200u 0 1e-8
let ac_ = v(mid) - mean(v(mid))
let mv = mean(ac_*ac_)
echo VAR $&mv
.endc
.end

measured var[v(mid)] = 0.000e+00      kT/C = 4.144e-12
```

Exactly zero — not "small". The node is driven only by the deterministic DC solution.

2. The generator: a fixed grid, not the timestep

The one thing a transient-noise implementation must not do is scale the noise sample by the adaptive timestep. A white source of density S sampled at interval Δt has per-sample deviation $\sqrt{S/(2\Delta t)}$. If Δt is the LTE-controlled simulation step, then every time the integrator changes step the injected power changes, and the resulting spectrum is an artefact of the step controller rather than a property of the circuit.

ngspice avoids this with a fixed noise grid. `struct trnoise_state (src/include/ngspice/1-f-code.h)` holds a sequence sampled at a period TS that the user chooses, independent of the simulation step:

```
struct trnoise_state {
    double points[TRNOISE_STATE_MEM_LEN];
    size_t top;
    double NA, TS, NAMP, NALPHA, RTSAM, RTSCAPT, RTSEMT;
    double *oneof;          /* pre-computed 1/f sequence, if NALPHA > 0 */
    size_t oneof_length;
    ...
};
```

NA is the white amplitude, NAMP/NALPHA the 1/f amplitude and exponent, and the RTS* parameters describe a random-telegraph burst. Samples are produced two at a time (rgauss returns a pair) by trnoise_state_gen, and any timepoint is served by trnoise_state_get, which generates forward until the requested index exists.

Interpolation between grid points

The device load functions do not hold the sample constant across a grid interval; they interpolate linearly (src/spicelib/devices/isrc/isrcload.c):

```
size_t n1 = (size_t)floor(time / TS);
double V1 = trnoise_state_get(state, ckt, n1);
double V2 = trnoise_state_get(state, ckt, n1 + 1);
value = V1 + (V2 - V1) * (time / TS - (double)n1);
```

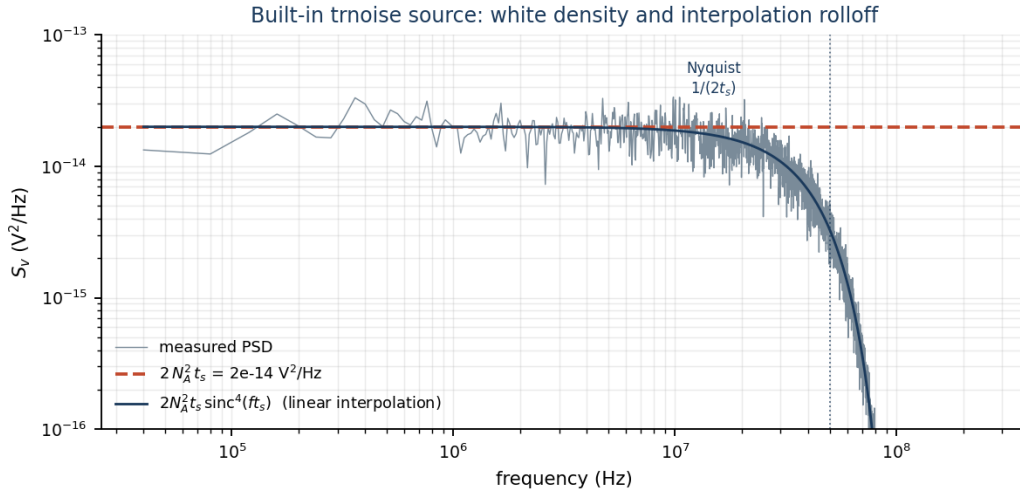
This has a spectral consequence that is worth deriving, because it is visible in every measurement. The continuous waveform is

$$x(t) = \sum_k V_k \Lambda\left(\frac{t}{t_s} - k\right)$$

with Λ the unit triangle of half-width t_s and V_k i.i.d. of variance N_A^2 . For a pulse-amplitude-modulated process with pulse h , $S_x(f) = \frac{\sigma^2}{t_s} |H(f)|^2$, and the triangle has $H(f) = t_s \text{sinc}^2(ft_s)$. Hence the one-sided density is

$$S_x(f) = 2 N_A^2 t_s \text{sinc}^4(ft_s)$$

which is flat at $2N_A^2 t_s$ for $f \ll 1/t_s$ and rolls off into Nyquist.



Measured with $N_A = 1$ mV, $t_s = 10$ ns, over a band of 990 bins well below Nyquist:

quantity	measured	theory	ratio
S_x (flat region)	1.929e-14 V ² /Hz	$2N_A^2 t_s = 2.0\text{e-}14$	0.965 ± 0.030

The measured curve follows the sinc^4 prediction through the entire rolloff, which is a stronger statement than matching the flat level: it confirms the interpolation model, not just the amplitude.

One consequence to be aware of when reading a raw waveform: because interpolation fills in between grid points, the variance of the sampled waveform is not N_A^2 . Integrating the density above gives $\frac{2}{3}N_A^2$ in the limit of dense sampling, and the simulator's own timepoints give something between that and N_A^2 depending on how they fall relative to the grid — measured 0.740 ± 0.003 here. The density is the well-defined quantity; the raw sample variance depends on where the timepoints land.

3. The OSDI path

A Verilog-A noise contribution reaches the simulator as a parametric description that the compiler emits and, before E-364, nothing read:

```
descr->noise_source_type[i]  /* NOISE_TYPE_WHITE / _FLICKER / _TABLE      */
descr->load_noise_params()    /* per-source `power` and `exponent`,          */
                             /*  $S(f) = \text{power} / f^{\text{exponent}}$ , at the      */
                             /* CURRENT bias                                */
descr->noise_sources[i].nodes /* the node pair the source injects between */
descr->noise_sources[i].name /* the correlation group (LRM 4.6.4)          */
```

so transient noise for OSDI devices is a pure simulator-side feature: no ABI change and no compiler change. `osdi_trnoise_stamp(src/osdi/osditrnoise.c)` is called from the OSDI load path and adds a current between the source's nodes.

Activation is automatic. Transient noise turns on when the deck already contains at least one `trnoise` independent source, and adopts that source's TS. The rationale is that such a deck is already running a noisy transient, and its OSDI devices were the only silent components in it. It is deliberately *not* keyed on "this device declares noise sources", because essentially every real compact model (BSIM, HICUM, MEXTRAM...) declares thermal and flicker noise, so that test would be equivalent to "always on" — making every existing transient result stochastic and costing convergence for a feature the deck never asked for. A deck with no transient-noise source is bit-identical to before.

Bias dependence. Device noise is not stationary: shot noise follows the current, flicker follows I^{AF} . power is therefore re-read every timepoint and the generator produces a unit-variance sequence that is scaled at load time. This is the standard quasi-stationary approximation — it assumes the operating point moves slowly compared with t_s , which is also the assumption that makes t_s meaningful.

Correlation. Verilog-A expresses perfectly correlated sources by giving them the same name (LRM 4.6.4). Correlated sources must therefore share one random stream rather than draw independently, so streams are keyed on the source name, not its index. power may be negative — OpenVAF folds the contribution factor as $\text{fac} \cdot |\text{fac}|$ so the sign carries direction ([Enhancement-42](#)) — and that sign is applied to the amplitude so correlated sources add coherently.

Scope. `white_noise()` and `flicker_noise()` are injected. `noise_table()` is not: a tabulated spectrum needs arbitrary frequency *shaping* rather than a scalar amplitude, which neither generator can express. It is skipped with a warning and remains fully accounted for in `.noise`.

4. The amplitude law, derived

ngspice's 1/f generator is Kasdin's fractional-noise method: white noise of deviation Q filtered by the FIR

$$h_k = h_{k-1} \frac{\alpha/2 + k - 1}{k}, \quad H(z) = (1 - z^{-1})^{-\alpha/2}$$

so that $|H(e^{j\omega})|^2 = (2 \sin(\omega/2))^{-\alpha} \rightarrow (2\pi f t_s)^{-\alpha}$ well below Nyquist. Combining with the discrete-white result of section 2, the generator delivers

$$S(f) = 2Q^2 t_s (2\pi f t_s)^{-\alpha}.$$

The model asks for $S(f) = |power|/f^\alpha$. Equating the two and solving:

$$Q = \sqrt{\frac{|power| (2\pi t_s)^\alpha}{2 t_s}}$$

which is exactly the line in `osdi_trnoise_stamp`:

```
double alpha = (descr->noise_source_type &&
    descr->noise_source_type[i] == NOISE_TYPE_FLICKER)
    ? ((exponent[i] > 0.0) ? exponent[i] : 1.0) : 0.0;
amp = sqrt(fabs(power[i]) * pow(2.0 * M_PI * ts, alpha) / (2.0 * ts));
```

Two properties matter. $\alpha = 0$ collapses to $\sqrt{|power|/(2t_s)}$, the white case, so both kinds share one expression. And neither depends on the run length, which is what makes results reproducible across `tstop`.

This law was *derived*, not fitted, and the derivation is what caught the original bug: before it, the white amplitude $\sqrt{|power|/(2t_s)}$ was being used for flicker too, which is wrong by $\sqrt{1/(2t_s\pi)}$ — a factor of 399 at $t_s = 1 \mu\text{s}$. The measurement returned $397\times$ and $396\times$, close enough to the predicted $398.9\times$ to confirm the derivation rather than invite a fudge factor.

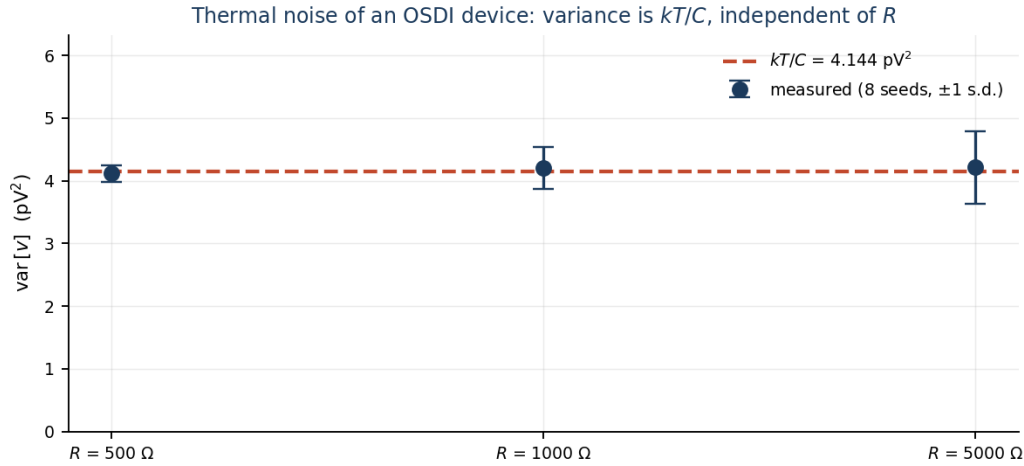
5. Linear device: thermal noise, and why kT/C is the strongest test

```
`include "disciplines.vams"
module valin(p, n);
    inout p, n;
    electrical p, n;
    parameter real r = 1e3 from (0:inf);
    parameter real c = 1e-9 from [0:inf);
    analog begin
        I(p, n) <+ V(p, n) / r;
        I(p, n) <+ ddt(c * V(p, n));
        I(p, n) <+ white_noise(4.0 * 1.3806488e-23 * $temperature / r, "thermal");
    end
endmodule
```

A white current source of density $4kT/R$ across a parallel RC gives

$$\text{var}[v] = \int_0^\infty \frac{4kT}{R} \frac{R^2}{1 + (2\pi f RC)^2} df = 4kTR \cdot \frac{1}{4RC} = \frac{kT}{C}.$$

R cancels. That is what makes this the strongest available check: it is parameter-free, so sweeping R over a decade and recovering the same variance tests the amplitude law rather than a coincidence of one parameter set. The deck uses a $1 \text{ G}\Omega$ series element so the VA device is the only resistance at the node (section 1: the built-in resistor would not contribute anyway, but this keeps the loading negligible too).



Eight seeds per point, $C = 1 \text{ nF}$, $kT/C = 4.144 \text{ pV}^2$:

R	measured var	ratio to kT/C
500Ω	$4.181\text{e-}12 \text{ V}^2$	1.009 ± 0.043
$1 \text{ k}\Omega$	$4.143\text{e-}12 \text{ V}^2$	1.000 ± 0.068
$5 \text{ k}\Omega$	$4.275\text{e-}12 \text{ V}^2$	1.032 ± 0.095

The variance is flat across a $10\times$ change in R , as the identity requires.

6. Nonlinear device: shot noise must track the bias

Thermal noise of a fixed resistor is a constant, so matching it only proves the amplitude scaling. Shot noise is $S_i = 2qI$ — it follows the operating point, so it also proves that `load_noise_params` is re-read at the current bias each timepoint rather than latched once at the DC solution.

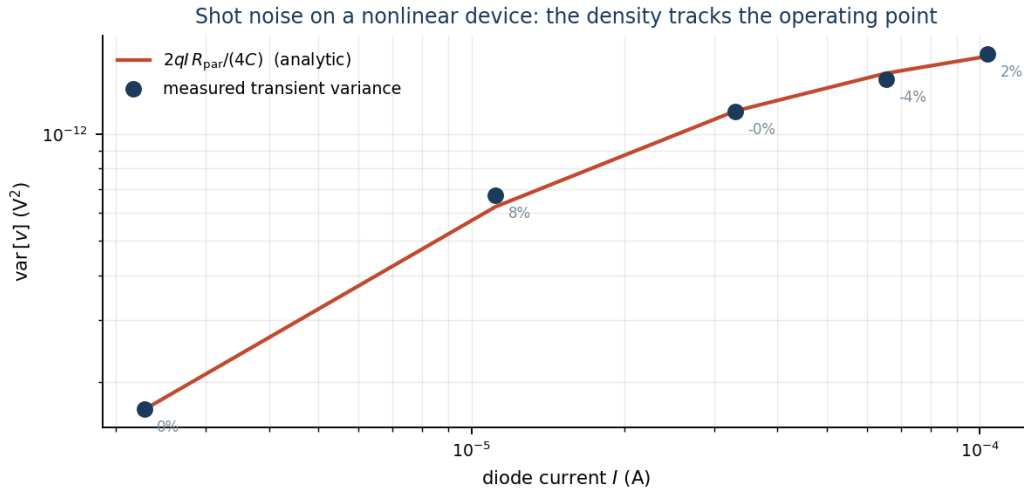
```

`include "disciplines.vams"
module vadio(a, c);
  inout a, c;
  electrical a, c;
  parameter real is = 1e-14 from (0:inf);
  parameter real nf = 1.0 from (0:inf);
  real vt, id;
  analog begin
    vt = 1.3806488e-23 * $temperature / 1.602176565e-19;
    id = is * (exp(V(a, c) / (nf * vt)) - 1.0);
    I(a, c) <+ id;
    I(a, c) <+ white_noise(2.0 * 1.602176565e-19 * abs(id), "shot");
  end
endmodule

```

For each bias the oracle is built from the diode's own operating point as reported by ngspice — $r_d = n_f V_T / I$, $R_{\text{par}} = r_d \parallel R_{\text{ext}}$, and the same integral as above:

$$\text{var}[v] = 2qI \frac{R_{\text{par}}}{4C}.$$



Eight seeds per bias, $R_{\text{ext}} = 1 \text{ k}\Omega$, $C = 1 \text{ nF}$:

V_{bias}	I	r_d	measured var	analytic	ratio
0.50 V	2.276 μA	11.36 $\text{k}\Omega$	1.705e-13	1.676e-13	1.018 ± 0.070
0.60 V	33.06 μA	782 Ω	1.140e-12	1.163e-12	0.980 ± 0.025
0.65 V	65.42 μA	395 Ω	1.421e-12	1.485e-12	0.957 ± 0.038
0.70 V	103.5 μA	250 Ω	1.619e-12	1.658e-12	0.976 ± 0.017

The current spans $45\times$ and r_d spans $45\times$ in the opposite direction; the agreement holds throughout. A latched-at-DC implementation would track the first row and drift on the rest.

The same sweep is available as a committed script, [examples/trnoise_examples/validate/psd_shot.py](#), which reports per-bias deviations of 2.3 %, 1.7 %, 1.0 % and 3.6 %.

7. Flicker noise — and a measurement trap worth documenting

```

`include "disciplines.vams"
module vaflick(p, n);
  inout p, n;
  electrical p, n;
  parameter real r = 1e3 from (0:inf);
  parameter real kf = 1e-20 from [0:inf);
  analog begin
    I(p, n) <+ V(p, n) / r;
    I(p, n) <+ flicker_noise(kf, 1.0, "fl"); //  $S_i(f) = kf / f$ 
  end
endmodule

```

With no capacitor the node impedance is a pure resistance, so $S_v(f) = k_f R_{\text{par}}^2 / f$ exactly — no pole to correct for.

Flicker noise follows $1/f$ at the predicted level

Measured against `.noise` on the same circuit (two independent code paths), 60 Welch segments over 153 bins:

quantity	result
mean ratio transient / .noise	0.9737 ± 0.0107
fitted PSD slope	-1.036 (ideal -1)

The trap

A first attempt at this measurement used $R = 1 \text{ k}\Omega$ instead of $1 \text{ M}\Omega$ and reported the flicker level 38 % low, consistently, across seeds — looking every bit like a real defect. It was not. Two things were wrong, and both are easy to repeat:

1. The reference was contaminated. In `.noise`, the external $1 \text{ k}\Omega$ resistor contributes $4kT/R_s$, which at that impedance is *larger* than the flicker term. The `.noise` output was $6.644 \times 10^{-18} \text{ V}^2/\text{Hz}$ at 1 kHz , of which 4.144×10^{-18} is the resistor and only 2.50×10^{-18} is flicker — and the resistor contributes nothing to the transient (section 1). The decomposition checks exactly: $6.644 - 4.144 = 2.500$, and $k_f R_{\text{par}}^2 / f = 2.50 \times 10^{-18}$ at 1 kHz .
2. The estimator was biased by the topology. At low impedance the flicker term is shallow against the floor, and windowing leakage flattens the measured slope (fitted -0.958 rather than -1). Evaluating that fit at the *edge* of the fit band then reads low.

At $1 \text{ M}\Omega$ the flicker term dominates the resistor's thermal contribution by $\sim 6000\times$, both problems vanish, and the ratio is 0.974 . The lesson is general: compare a transient measurement with `.noise` only in a regime where the noise you are measuring dominates the sources the two analyses treat differently.

8. Cross-check against `.noise`

The sharpest end-to-end test compares variances rather than a single band, because integrating the `.noise` spectrum uses every data point from both analyses and avoids the narrow-band scatter discussed in section 9:

```
noise v(mid) V1 dec 50 1e2 1e9    → integrate S_v df    → 4.143e-12 V2
tran 1e-8 600u 0 1e-8             → var[v(mid)]         → 4.113e-12 V2
```

quantity	value
variance from integrating <code>.noise</code>	$4.143\text{e-}12 \text{ V}^2$
variance measured in <code>.tran</code> (8 seeds)	$4.113\text{e-}12 \pm 2.2\text{e-}13 \text{ V}^2$
ratio	0.993 ± 0.052
kT/C (independent closed form)	$4.144\text{e-}12 \text{ V}^2$

Three independent routes — a frequency-domain small-signal analysis, a time-domain stochastic simulation, and a closed-form thermodynamic identity — agree to well within the statistical error.

9. How to measure this correctly

A Welch PSD estimate is unbiased, but its relative scatter over a band of M bins averaged over S segments is $\approx 1/\sqrt{MS}$. Checked against a synthetic i.i.d. sequence of *known* density ($2\text{e-}14 \text{ V}^2/\text{Hz}$):

band	bins	estimate
wide, high-frequency	~ 1000	$1.96\text{--}2.01\text{e-}14$ ($\pm 2 \%$)
narrow, low-frequency	~ 10	$1.53\text{--}2.46\text{e-}14$ ($\pm 25 \%$)

Reading an amplitude off a narrow band therefore says more about the estimator than about the simulator. The practical rules used throughout this note:

- Amplitude $\rightarrow$ variance. It integrates every bin and is tight. Average over independent seeds (setseed) and quote a standard deviation.
- Shape $\rightarrow$ PSD. Flatness and the 1/f slope are robust to bin scatter.
- Flicker is the exception. A 1/f process has no stationary variance (it depends on the run length), so its level must come from a wide-band fit or a matched comparison against .noise.
- Validate the instrument first. Every clean result in this note was preceded by running the estimator on a signal whose answer was already known.

10. Practical notes

Choosing t_s . It sets the noise bandwidth: content exists up to $1/(2t_s)$, shaped by sinc^4 . Choose it a few times faster than the fastest circuit time constant you care about, and no faster — halving t_s doubles the sample count and, for flicker, the memory.

Memory. A 1/f generator pre-computes its whole sequence for the run (f_alpha is called once with CKTfinalTime/TS points), costing ~8 bytes per source per instance per noise timestep. White sources stream and cost nothing. A long transient with many flicker sources is the case to watch.

Timestep. Use a maximum-step argument (tran <step> <stop> 0 <step>). Without it the integrator will happily take steps far longer than t_s through quiet intervals and alias the noise.

$t = 0$. No noise is injected at $t = 0$; the operating point must be the deterministic DC solution, or the starting point of every run would move.

Reproducibility. setseed <n> before tran fixes the stream, which is what makes the seed-averaged statistics above possible.

11. Known limits

- noise_table() is not injected in transient (warned, and still correct in .noise).
- Built-in devices remain noiseless in .tran; only trnoise sources and OSDI devices inject. Model a noisy passive as an OSDI device if you need it.
- The quasi-stationary approximation assumes the bias moves slowly compared with t_s . A circuit slewing significantly within one noise interval is outside what this scheme represents.
- Under USE_OMP the OSDI evaluation path has known thread-safety problems affecting @(initial_step) and \$strobe; the shipped binary is built without it.

Reproducing everything here

```
python3 docs/internals/ngspice_internals/make_trnoise_figs.py
python3 examples/trnoise_examples/verify_trnoise.py
python3 examples/trnoise_examples/validate/psd_shot.py
python3 examples/trnoise_examples/validate/psd_flicker.py
```

The first regenerates all four figures (pass white, ktc, shot or flicker to do just one). The second is the fast regression check. The last two are the statistical measurements that established the amplitude law; they take minutes and need numpy.

References

- [Enhancement-364](#) — transient noise for OSDI devices
- [Enhancement-42](#) — correlated noise sources and the sign convention
- src/frontend/trannoise/1-f-code.c — f_alpha, trnoise_state_gen
- src/osdi/osditrnoise.c — the OSDI injection path

- N. J. Kasdin, "Discrete simulation of colored noise and stochastic processes and $1/f^\alpha$ power law noise generation", *Proc. IEEE* 83(5), 1995